ANDREA DI FRANCIA

Data Scientist with 8+ years of expertise in public and private sector roles, utilizing statistical and machine learning models, including LLMs and Gen-AI solutions, to derive insights and build data-driven products. Skilled in Python, R, SQL, extracting insights from unstructured data, and productionising ML models leveraging Docker and Airflow.

andrea.l.difrancia@gmail.com | andreadifra | andreadifrancia | London, UK

EXPERIENCE

Data/Machine Learning Scientist (Contractor)
Bupa
Nov 2025 – Present | Remote, UK

- Delivered **health assessment outcomes analysis** on **107,000+ patient records** in **Snowflake**, supporting an upcoming **academic white paper** quantifying impact on customer health.
- Led initial development of **clinical risk models** and prototype **Markov cost-effectiveness model**, directly informing Bupa's **AI personalisation** strategy.
- Designed **production roadmap** for AI-powered **Personalised Health Plans** platform, defining **MLOps architecture**, clinical governance, and deployment strategy on **Azure/Snowflake** for integration into the Blua patient app.

Lead Data Science Consultant
Health Economics and Outcomes Research (HEOR)
Oct 2024 – Nov 2025 | Remote, UK

- Developed **mixture regression models** to predict 3-year healthcare costs from obesity-related complications, quantifying **£3,000+ savings per patient** from reducing risk of type 2 diabetes.
- Conducted **real-world evidence analysis** on **161,000+ T2D patients** using **parametric survival models**, quantifying higher CVD risk (10/15% increase) for high-BMI patients over 10 years.
- Led **global meta-analysis** of meningococcal carriage prevalence using **hierarchical mixed-effects models** to inform vaccination strategies across regions and age groups.

Senior Data Scientist
UK Health Security Agency
Jan 2022 – Sep 2024 | London, UK

- Developed **Text Classification models** using **Transformer-based LLMs** to extract hepatitis markers from unstructured clinical notes (**95%+ accuracy**), reducing manual review by **10 hrs/month**.
- Implemented a **Bayesian model** (***R/Stan***) estimating the NHS Covid-19 App reduced cases by ~**1m**, **40k hospitalisations** & **3k deaths**; partnered with **University of Oxford** on **published research**.
- Leveraged cloud platforms (**AWS, Azure**) with **Docker** and **Airflow** to productionise analysis at scale, including **ETL pipelines** and **non-linear regression models** on genomic sequencing data.

Economic Data Scientist
Department for Education
Apr 2017 – Dec 2021 | London, UK

- Developed ***R***-based **forecasting models** (**£24bn** teacher pay-bill) and **cost pressures models** (**£40bn** schools expenditure), with interactive ***shiny*** dashboards supporting policy development.
- Quantified economic benefits of successful **£240m budget proposal** for the Early Career Framework policy and created ***R-shiny*** dashboard tracking key performance metrics that received an award for outstanding achievement.
- Led analysis on capital expenditure and financial sustainability for the Higher Education sector in England (**£4.3bn**), quantifying market failure risks and rationale for government intervention.

SKILLSET

Python | SQL | R | Snowflake | Azure | Survival Analysis | Health Economics | Epidemiology | LLMs | NLP | Docker | Mixed-effects Models | Bayesian Methods | Deep Learning | PowerBI | Airflow | Mlflow | Clustering | Time Series | PySpark | PyTorch

EDUCATION

MSc. Applied Statistics & Operational Research
Birkbeck, University of London
Grade: **Distinction**

- Top 10% in cohort; highest marks: Stochastic Systems (94%), Statistical Learning (87%)
- 2-year part-time study programme; completed whilst working full-time

BSc. (Hons) Economics
University of Nottingham
Grade: **1st Class Honours**

PUBLICATIONS

Journal Articles

- L. Ferretti et al., "Digital measurement of SARS-CoV-2 transmission risk for precision epidemiology," *Nature*, 2023. [Online]. Available: https://rdcu.be/dvLyt
- M. Kendall et al., "Epidemiological impacts of the NHS COVID-19 app in England and Wales throughout its first year," *Nature Communications*, vol. 14, 2023. [Online]. Available: https://rdcu.be/c6yas

INTERESTS

Computer Vision
Spearheaded a hackathon project focused on automating the extraction of tables and handwritten text from a backlog of source documents containing disease notifications. Leveraged Tika and Tesseract for robust object recognition on images, facilitating data extraction for subsequent analysis and integration into downstream applications.

Mathematics
Completed half of the required modules (60 UK CATS) towards the Graduate Diploma in Mathematics offered by the University of London (academic direction from London School of Economics).